Cover Letter

November 1, 2026

Faculty Search Committee
Electrical and Computer Engineering
School of Engineering, Penn State Behrend
Erie, Pennsylvania

Re: Tenure-Track Assistant Professor in Electrical or Computer Engineering, REQ_0000072782

Dear Members of the Search Committee:

I am writing to apply for the tenure-track Assistant Professor position in Electrical or Computer Engineering at Penn State Behrend. I earned my PhD in Electrical and Computer Engineering from the University of Oklahoma, where I investigated AI/ML methods for wireless network reliability, propagation modeling, and adaptive optimization. I am interested in joining a department where engineering teaching, applied scholarship, and industry partnerships reinforce each other, particularly through meaningful involvement of undergraduate students.

My research asks how intelligent systems can make dependable predictions and decisions when measurements are scarce or conditions change. In my doctoral work, I studied data-efficient radio propagation models, generative augmentation, domain-aware transfer learning, and methods for detecting and compensating for network failures. These projects have led to peer-reviewed publications in communications and sensing venues and provide a foundation for a broader program in reliable learning for networked and sensor-based systems. Related work on smartwatch inertial signals extends these questions to time-series recognition and cross-user variability.

At Behrend, I would build a focused research program around three connected questions: how to transfer or augment models between different engineering environments without making predictions worse; how uncertainty-aware monitoring can support resilient network decisions; and how lightweight learning methods can handle unreliable sensor measurements. The initial work can be accomplished with carefully documented simulation and existing datasets, allowing students to reproduce a baseline before developing a new method. Over time, I would seek collaborations with faculty working in sensing, embedded and cyber-physical systems, communications, and applied AI, and pursue appropriate external research support. My experience at AT&T Labs, Futurewei, and AT&T gives me practical perspective on industrial constraints and collaboration, without presuming access to private company data or a current research grant.

I also bring university teaching and mentoring experience. At the University of Lahore, I taught Computer Programming, Communication Systems, and Wireless Communication to undergraduates in a Technology department. At UET Taxila, I supported communications laboratories and course assessment, and at COMSATS I mentored engineering research students. My approach connects a mathematical model to a reproducible experiment, then asks students to explain its limits. My strongest instructional alignment is in communications, programming, signal and data analysis, wireless systems and related laboratories. I would prepare carefully for other courses within the EE, CE and ECET programs rather than claim prior expertise teaching every topic in the posting.

Penn State Behrend's undergraduate research and industry-connected engineering environment is particularly attractive to me. I would involve students in projects that begin with an interpretable baseline and a clearly bounded test, and develop toward senior design, presentations, and publishable findings where warranted. I would also contribute to curriculum, advising and service with the same attention to preparation and clear communication that I bring to technical work.

Thank you for considering my application. I would welcome the opportunity to discuss my research plans, teaching contributions, and how they can support Penn State Behrend's engineering students and growing applied research programs.

Sincerely,

Waseem Raza, PhD

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